

GENERAL INTRODUCTION

The philosophy of science is at least as old as Aristotle, but it has risen to special prominence since the beginning of the twentieth century. As scientists have made tremendous advances in fields as diverse as genetics, geology, and quantum mechanics, increasing numbers of philosophers have made science their focus of study. In its broadest terms, the philosophy of science is the investigation of philosophical questions that arise from reflecting on science. What makes these questions philosophical is their generality, their fundamental character, and their resistance to solution by empirical disciplines such as history, sociology, and psychology.

Contrasting different sorts of questions can bring out the difference between the philosophy of science and other disciplines that study science. For example, "When was the planet Neptune discovered?" is primarily a question for historians, not for philosophers.¹ Similarly, "Why did Soviet biologists under Stalin reject Mendelian genetics?" or "Why did James Watson underrate the contributions of Rosalind Franklin to the work that led to the discovery of the double helix structure of DNA?" fall within the domains of sociology, political science, and psychology. Contrast these questions with the following: "When is a theory confirmed by its predictions?" "Should we be realists about all aspects of well-established theories?" "What is a law of nature?" These questions are philosophical. They cannot be answered simply by finding out what has happened in the past or what people now believe.

For similar reasons, philosophical questions about science cannot be answered by the sciences themselves (although being able to answer these questions often depends on having a good understanding of scientific theories). A geneticist at the National Cancer Institute, for example, might ask whether certain people are born with a natural immunity to AIDS and set out to answer this question through empirical research. But if our geneticist asked "What is a law of nature?" or "What is science?" or "When is a theory confirmed?" she would not discover the answer by doing more science.

The central questions in the philosophy of science do not belong to science as such; they are about science, but not part of it. Of course, scientists can be (and sometimes have been) philosophers of science. The point is that when people are doing philosophy of science, they are not (usually) doing science per se, and most philosophers of science (at least since the twentieth century) have not been practicing scientists. Thus, the philosophy of science is not a branch of science but belongs to philosophy, and it intersects with other areas of philosophy, such as epistemology, metaphysics, and the philosophy of language.

The aim of *Philosophy of Science: The Central Issues* is to introduce the reader to the main currents in modern philosophy of science. It is primarily intended for use in introductory courses at both the undergraduate and

graduate levels. In order to keep the book within manageable bounds, some difficult decisions had to be made about what to include and what to exclude. In making these decisions we were guided by our own experience in teaching at Purdue and the recommendations of our reviewers who contributed significantly to the book's development.

The first and in some ways the easiest decision was to exclude almost entirely the social sciences and concentrate on the natural sciences.* In this we followed the lead of other texts. The philosophical questions raised by disciplines such as history, psychology, sociology, and anthropology are fascinating and important. But they are so different from the questions one encounters in physics, biology, and chemistry that they would require another volume, comparable in length to this one, in order to address them adequately.

A second decision, which we made at the outset, was to avoid foundational questions about the concepts, structure, and content of particular theories and to focus instead on general issues that arise across scientific disciplines. Thus, this volume is organized around wide-ranging philosophical topics and problems, not individual theories or sciences. Details of particular sciences are introduced rarely and only when necessary for evaluating a philosophical position or argument (as, for example, in the chapter on reduction). In this way we hope to avoid the trap of turning a philosophy of science course into a mimic course in science and to keep the focus on the philosophy in the philosophy of science. It also has the advantage of making courses based on this book accessible to students (even those at the graduate level) whose background in the sciences may be slight or nonexistent. For the same reason of accessibility, we have confined our selections to readings that use no more than a bare minimum of logical or mathematical notation.

Our approach, then, is focused on philosophical topics and problems, not on particular sciences and theories. A consequence of this topics-and-problems approach is that the chapters of *Philosophy of Science: The Central Issues* pay little attention to tracing the historical development of the philosophy of science. Although we devote some time to filling in some of the essential historical background in the commentaries, this volume is not historical in the way that it treats ideas, arguments, or philosophers. What connects the readings (and the discussions of them in the commentaries) is their focus on common themes, arguments, and criticisms, regardless of whether the authors share the same nationality, are writing in the same decade, or belong to the same school. Our approach is not antihistorical, but it is largely ahistorical.

Because of the many sharp disagreements within the philosophy of science and the unresolved character of nearly all the fundamental questions

* One notable exception is our inclusion in chapter 8 of Jerry Fodor's influential article defending the autonomy of the special sciences (which include sociology, psychology, and economics).

that philosophers ask about science, in an anthology seemed to us to be the only sane choice for a book intended for use in the classroom. But the anthology format brings with it a problem that just about every teacher of philosophy of science has had to confront. Hardly any of the readings, whether old classics or brand-new articles, were written with students in mind. Rather, they were published in books and professional journals, addressed primarily to fellow professionals. Thus, they often presuppose an awareness of issues, positions, and arguments, both in the philosophy of science and in philosophy more generally, that most students lack. Consequently, even the brightest students can find it hard to understand the material they are being asked to read, discuss, and evaluate. The most common complaint voiced by the teachers we spoke with in the several years that went into planning and writing this book is that many of the readings in the existing anthologies are too sophisticated—they make too many references to the history of science and allude too frequently to philosophical ideas and arguments for the beginning student to get much out of them. What was needed, and what we have tried to provide here, is a serious, comprehensive guide that will really help students in their first encounter with the readings. Thus, in addition to short introductions to each chapter, we have written extended and often detailed commentaries on the readings. Getting the tone and level of detail right in these commentaries has been the hardest and most rewarding part of the book's development. Much of the fine tuning and, in some cases, the inclusion and deletion of entire sections, was guided by our reviewers. We have strived to make each commentary and the sections within them self-contained so that each can be used independently of the rest. And in order to maximize the pedagogical usefulness of *Philosophy of Science: The Central Issues*, each reading is linked explicitly with one or more of the sections into which the commentaries are divided. In this way, where one should look in the commentaries for discussion, explanation, background, and analysis of any of the fifty-two separate readings in the book should be clear.

At the end of this volume there is a glossary, a bibliography, and indexes of names and subjects. The glossary is comprehensive: it covers most of the terms that may be new to the reader or that are being used in an unfamiliar way. The bibliography is divided into nine sections, one for each chapter. Inevitably, this involves some repetition of titles of books and articles, but our aim was to provide the reader with suggestions for further reading, at an appropriate level, about the issues discussed in each chapter's commentary. Consequently, not everything cited in the commentaries appears in the bibliography, some items appear in the bibliography more than once, and there are some entries in the bibliography that are not mentioned in the commentaries.

The difference between an anthology and a heap of articles lies in their organization. But any system of division will be, to some extent, artificial and misleading: artificial because of the interrelated character of the issues

in the philosophy of science and misleading because it might suggest that the readings in one chapter are not connected with those in another. Thus, as with any collection of this kind, the reader or teacher needs to bear in mind that not everything pertinent to, say, the topic of laws will be found in the chapter devoted to laws and that relevant readings and commentaries might also appear in the chapters on induction, confirmation, and reduction (as indeed, in this case, they do). Moreover, this is a collection of readings on related topics, not an extended narrative with a beginning, a middle, and an end. Users of the book should not feel constrained by the order of the chapters or even, in most cases, by the order of the readings within those chapters, when deciding what to read first, what to read second, and so on. Obviously, we have arranged the material in an order that makes sense to us, trying wherever possible to juxtapose readings that speak to the same or closely related issues, but many different arrangements are possible and may be preferable, depending on one's interests and teaching goals.

In articles referring to material reprinted in this volume, and in all footnotes, internal page numbers are enclosed in square brackets. In the commentaries, internal page numbers appear in parentheses.

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1. This is not to deny that the question might raise philosophical issues concerning the concept of discovery. For example, while Galileo was observing Jupiter and its moons with his telescope, he saw Neptune and recorded its position on at least two occasions. Up to that time, no one had seen Neptune (because it is not visible to the naked eye). Galileo apparently regarded what he saw as a fixed star, not a planet (though he did note that the object appeared to have moved). Did Galileo discover Neptune some 234 years before its official discovery in 1846? See Stillman Drake and Charles T. Kowal, "Galileo's Sighting of Neptune," *Scientific American* 243 (1980): 52-59. For further discussion of the concept of discovery and its application to theoretical entities such as the electron, see Peter Achinstein, "Who Really Discovered the Electron?" in Jed Buchwald and Andrew Warwick, eds., *Histories of the Electron: The Birth of Microphysics* (Cambridge, Mass.: MIT Press, 2001), reprinted in Peter Achinstein, *Evidence, Explanation, and Realism* (New York: Oxford University Press, 2010), and Theodore Arabatzis, *Representing Electrons: A Biographical Approach to Theoretical Entities* (Chicago: University of Chicago Press, 2006).

| Philosophy of Science

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